

Statistics 651

Multivariate Models

Winter 2026

Objectives

1. Work with common multivariate distributions
2. Build (conditionally) conjugate models for multinomial and Multivariate Normal data

Resources

- BDA3:
 - Ch 3.4 (Multinomial, Dirichlet)
 - Ch 3.5-3.6 (Multivariate Normal)
- Hoff: Ch 7 (Multivariate Normal)
- [Rencher and Schaalje](#), Ch 4 (Multivariate Normal theory)

Categorical data

Multinomial distribution

Definition

Let $\mathbf{y} = (y_1, \dots, y_K) \in \{1, \dots, n\}^K$ be a vector of counts resulting from

- n independent trials, where
- each trial can result in category $1, \dots, K$, with
- probabilities given in $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)$, such that
- $\theta_k \geq 0$ and $\sum_{k=1}^K \theta_k = 1$.

Then the joint distribution of $\mathbf{y}$ is **multinomial** with probability mass

$$p(\mathbf{y} \mid \boldsymbol{\theta}) = \frac{n!}{\prod_{k=1}^K y_k!} \prod_{k=1}^K \theta_k^{y_k}.$$

It is denoted as $\mathbf{y} \sim \text{Multinom}(n, \boldsymbol{\theta})$.

Note: Marginal distribution of y_k is $\text{Binom}(n, \theta_k)$.

Dirichlet distribution

Definition

Let $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K) \in S^K$ where $S^K := \{\boldsymbol{\theta}_k \in [0, 1]^K : \sum_{j=1}^K \theta_j = 1\}$, and

let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$ with $\alpha_k > 0$ for all $k = 1, \dots, K$.

Then $\boldsymbol{\theta}$ follows a **Dirichlet** distribution with shape parameters in $\boldsymbol{\alpha}$ if it has density

$$\pi(\boldsymbol{\theta} \mid \boldsymbol{\alpha}) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \theta_k^{\alpha_k - 1}.$$

It is denoted as $\boldsymbol{\theta} \sim \text{Dir}(\boldsymbol{\alpha})$.

What is this distribution when $K = 2$?

$$\frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} \theta_1^{\alpha_1 - 1} \theta_2^{\alpha_2 - 1}$$

Beta(α_1, α_2)!

Dirichlet distribution

Properties

Let $\boldsymbol{\theta} \sim \text{Dir}(\boldsymbol{\alpha})$ and let $\alpha_0 = \sum_{k=1}^K \alpha_k$. Then,

- $\mathbb{E}(\boldsymbol{\theta}) = (\mu_1, \dots, \mu_K) = \left(\frac{\alpha_1}{\alpha_0}, \dots, \frac{\alpha_K}{\alpha_0} \right)$, $\in \mathcal{S}^K$
- $\text{Cov}(\theta_i, \theta_j) = \frac{\mu_i I\{i=j\} - \mu_i \mu_j}{1 + \alpha_0}$,
- The mode is $\left(\frac{\alpha_1 - 1}{\alpha_0 - K}, \dots, \frac{\alpha_K - 1}{\alpha_0 - K} \right)$ for $\alpha_k > 1$,
- The marginal distributions are $\theta_k \sim \text{Beta}(\alpha_k, \sum_{j \neq k} \alpha_j)$,
- Distributions of subvectors of $\boldsymbol{\theta}$ collapse similarly.

See [Wikipedia's Dirichlet distribution](#) page for more properties.

Dirichlet distribution

Reparameterization

It is sometimes convenient to reparameterize to use

- **concentration** parameter $\alpha_0 := \sum_k \alpha_k$, and
- **mean** parameter $\boldsymbol{\mu} := \left(\frac{\alpha_1}{\alpha_0}, \dots, \frac{\alpha_K}{\alpha_0} \right)$,

so that $\boldsymbol{\alpha} = \alpha_0 \boldsymbol{\mu}$.

Definition

If all $\alpha_k = \alpha$ are equal so that

$$\alpha_0 = K\alpha,$$

$$\mu_1 = \mu_2 = \dots = \mu_K = 1/K,$$

we call the resulting distribution **symmetric Dirichlet**, written $\boldsymbol{\theta} \sim \text{Dir}(\alpha)$.

Dirichlet distribution

Constructive definition

Let $X_k \stackrel{\text{ind}}{\sim} \text{Gamma}(\alpha_k, 1)$ independently for $k = 1, \dots, K$.

Then $\left(\frac{X_1}{\sum_k X_k}, \dots, \frac{X_K}{\sum_k X_k} \right) \sim \text{Dir}(\alpha_1, \dots, \alpha_K)$.

How can this be useful?

- Show properties of Dirichlet
- * Draws from Dirichlet distribution

Dirichlet distribution



Constructive definition

Write an R function to generate a random draw from a Dirichlet distribution.

Multinomial-Dirichlet distribution

Marginal distribution

Suppose

$$\begin{aligned}\boldsymbol{\theta} \mid \boldsymbol{\alpha} &\sim \text{Dir}(\boldsymbol{\alpha}), \\ \mathbf{y} \mid \boldsymbol{\theta} &\sim \text{Multinom}(n, \boldsymbol{\theta}).\end{aligned}$$

Find the marginal distribution of $\mathbf{y}$.

$$\begin{aligned}P(\mathbf{y}) &= \int P(\mathbf{y} \mid \boldsymbol{\theta}) d\boldsymbol{\theta} = \int \underbrace{\text{Multinom}(\mathbf{y} \mid \boldsymbol{\theta})}_{\text{pmf}} \underbrace{\text{Dir}(\boldsymbol{\theta} \mid \boldsymbol{\alpha})}_{\text{density}} d\boldsymbol{\theta} \\ &= \int \frac{n!}{\prod_k y_k!} \prod_k \theta_k^{y_k} \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_k \theta_k^{\alpha_k - 1} d\boldsymbol{\theta} \\ &= \frac{n!}{\prod_k y_k!} \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \int \prod_k \theta_k^{\alpha_k + y_k - 1} d\boldsymbol{\theta}\end{aligned}$$

$$= \frac{n!}{\prod y_k!} \frac{\prod_k \Gamma(\alpha_k)}{\prod_k \Gamma(\alpha'_k)} \frac{\prod_k \Gamma(\alpha'_k)}{\Gamma(\sum \alpha'_k)}$$

where
 $\alpha'_k = \alpha_k + y_k$

Dirichlet distribution

Conjugacy

Show that the Dirichlet distribution is conjugate to a multinomial likelihood.

$$P(\underline{\theta} | \underline{y}) = \frac{P(\underline{\theta}, \underline{y})}{P(\underline{y})} = \frac{\text{Multinomial}(\underline{y} | \underline{\theta}) \text{Dir}(\underline{\theta} | \underline{\alpha})}{\int \text{above } d\underline{\theta}}$$

$$= \text{Dir}(\underline{\theta} | \underline{\alpha}')$$

where $\underline{\alpha}' = \underline{\alpha} + \underline{y}$

That is,

$$\underline{\theta} | \underline{y} \sim \text{Dir}(\underline{\alpha}')$$

Multinomial modeling

Example

Perform an analysis of the M&M data.

- Summarize the posterior distribution of θ under common priors.
- Are red M&Ms more likely than blue?
- Are colors of M&Ms evenly distributed?